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Multi-Stage Classification of Parkinson's Disease Using Functional Brain Networks and Machine Learning

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São Carlos – SP

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To all the giants whose shoulders I stand on

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“As invenções são, sobretudo,
o resultado de um trabalho de teimoso.”
(Santos Dumont)

ABSTRACT

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The clinical diagnosis of Parkinson's Disease (PD) is notoriously unreliable, with accuracy in early-stage, untreated cases as low as 26%. This diagnostic gap highlights an urgent need for quantitative, interpretable biomarkers. This thesis develops and validates a multi-stage (Control, Prodromal, PD) classification framework by integrating resting-state fMRI (rs-fMRI) connectomics with machine learning. We systematically compared two feature-engineering approaches: (i) network-based graph metrics, and (ii) high-dimensional region-to-region functional connectivity (FC). Our results demonstrate a clear distinction: the graph-metrics approach performed poorly (best test F1-macro = 0.478). In contrast, the direct FC approach proved far superior, with a Logistic Regression classifier using the functionally-derived Shen 268 atlas achieving a test F1-macro score of 0.665 (a 39% relative improvement). Beyond classification, we identified interpretable neurobiological markers. Both statistical group analysis and SHAP (SHapley Additive exPlanations) analysis converged, revealing that the model's predictions were driven by widespread decreases in PD brains connectivity. These disruptions were centered on the Subcortical-Cerebellar (SC) network, identifying both a loss of internal connectivity within the SC network and a disconnection between the SC network and other cortical systems, such as the Motor network. This research validates the superiority of raw FC data over abstracted metrics, provides quantitative support for PD as a "disconnection syndrome," and establishes a robust, explainable framework to aid in the diagnosis of Parkinson's disease.

Key-words: Parkinson, Brain Connectomics, Machine Learning, Complex Networks, fMRI.

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Chapter 1

INTRODUCTION

1.1 Motivation and Contextualisation

According to the World Health Organization (WHO) 1, Parkinson's disease affects 1 percent of the world's population over the age of 65, making it the second most common neurodegenerative disease in the world, behind Alzheimer's disease. The diagnosis of Parkinson's disease is usually clinical, based on social behaviour, motor symptoms of tremor, involuntary movements, slow movements and rigidity, as well as non-motor symptoms such as dementia and cognitive impairment (SVEINBJORNSDOTTIR, 2016).

The overall validity of clinical diagnosis of Parkinson Disease (PD) is unsatisfactory, according to Rizzo *et al.* (2016). In a review carried out on 20 published studies, the author reported an accuracy of around 80%, with diagnostic efficiency not increasing significantly in the last 25 years. At least 15% of patients with a diagnosis of Parkinson's disease in the population do not fulfil strict clinical criteria for the disease (SCHRAG; BEN-SHLOMO; QUINN, 2002).

PD diagnosis in the early stages remains highly inadequate. All current clinic diagnostic criteria consider signs and symptoms emerging in the later stage of disease. Studies like (ADLER *et al.*, 2014) mention staggering low accuracy values, specially in untreated or not clearly responsive subjects, where the accuracy can be as low as 26%.

The false diagnosis of Parkinson's disease in patients can affect the family and the patient's mental and physical well-being, increasing stigma and discrimination, as well as the risk of developing mental illness. For early PD patients, a false diagnosis can lead to lack of treatment and the rapid progression of the disease (PAHWA; LYONS, 2014).

These issues highlight the urgent need for quantitative, reproducible, and interpretable methods that (i) minimize dependence on subjective medical judgment, (ii) assess disease stage objectively, and (iii) provide explainable insights to assist physicians and medical institutions, especially in early diagnosis.

Previous studies (GöTTLICH *et al.*, 2013) suggest that Parkinson is a manifestation of changes in the brain organization and connectivity, a disconnection syndrome between functionally specialized brain regions which highlights less efficient Global network properties in PD patients when compared to healthy controls. Subsequent research (SANG *et al.*, 2015) further demonstrated that even patients in the early stages of PD exhibit altered topological brain

network with rs-fMRI (Resting State Functional Magnetic Resonance Imaging) data. These findings demonstrate that the structural organization of the brains of PD patients, across different stages of the disease, differs significantly from the intrinsic organization observed in healthy individuals.

Other papers like (KAZEMINEJAD; GOLBABAEI; SOLTANIAN-ZADEH, 2017) and (JIJI; RAJESH; LAKSHMI, 2023) have already used Complex Networks Theory and Machine Learning with fMRI data to diagnose PD patients. Nevertheless, few studies have conducted comparative analyses between correlation metrics and machine learning models, or have explored multistage classification with medical interpretation of how brain connectivity evolves throughout neurodegeneration. This gap motivates the present research.

1.2 Objectives

This project aims to develop an automatic diagnostic methodology for Parkinson's disease (PD) based on the integration of Machine Learning and Complex Network Theory, providing a computer-aided tool to support medical decision-making for patients who exhibit symptoms or early signs of PD.

Specifically, the research seeks to automatically diagnose PD across its different stages by employing functional Magnetic Resonance Imaging (fMRI) data and analyzing the topological properties of brain networks. Through this approach, we intend to identify and characterize the primary structural and functional differences between the brains of Parkinson's patients and those of healthy controls. These differences will be quantified through complex network measures, feature importance analyses and explainable AI methods like SHAP and LIME, enabling a deeper understanding of how PD affects brain organization.

Ultimately, this work intends to demonstrate that it is possible to accurately and interpretably classify PD patients using Machine Learning techniques. The final goal is to establish a quantitative, reproducible, and explainable diagnostic framework, contributing both to clinical support and to the broader understanding of the neurodegenerative progression of Parkinson's disease.

To achieve this objective, we will undertake the following activities.

1. Determinate the best method to preprocess the fMRI data into brain regions time series so that the adverse influences mentioned in [Parkes et al. (2018)], like head motion and fluctuations in non-neuronal physiological processes, are minimized .

2. Create a classification model involving Machine Learning and/or Deep Learning and validate its effectiveness in differentiating healthy patients from Parkinson's patients.

3. Differentiate the structure and relationship between brain regions of PD patients, early-stage PD patients and healthy controls, also giving medical interpretation to the brain network topology.

Chapter 2

FUNDAMENTAL CONCEPTS

2.1 Functional Connectivity and rs-fMRI in Neurodegenerative Diseases

At the microscopic level, a neuron consists of a cell body (soma) that receives input through dendrites and transmits electrical impulses through axons to other cells. These microscopic processes result in a localized increase in cerebral blood flow that exceeds the oxygen demands of neural activity, producing a measurable change known as the Blood Oxygenation Level–Dependent (BOLD) signal.

It is crucial to appreciate that the Functional Magnetic Resonance Imaging (fMRI) measures this BOLD response, which is a secondary and indirect measure of neuronal activity. Although indirect, fMRI BOLD signal responses are directly proportional to average neuronal firing rates (HEEGER *et al.*, 2000), making fMRI a valuable tool for studying brain activity and function.

Over the last two decades, fMRI data have emerged as a key method to reveal how components of large-scale distributed neural systems are coupled together in performing specific functions. The organization, interrelationship and integrated performance of these different regions are generally described by the term "functional connectivity" (FC). The FC approach examines statistical dependencies between spatially distinct brain regions, under the assumption that regions with correlated BOLD signal fluctuations are functionally connected (BIJSTERBOSCH; SMITH; BECKMANN, 2017). It is important to note that functional connectivity does not necessarily indicate direct anatomical links or causal interactions between brain regions. Instead, it reflects statistical correlations in activity that may arise from both direct and indirect neural pathways or shared modulatory influences. This framework enables the study of large-scale brain networks that support perception, cognition, and behavior.

In order to study connectivity, we often look at spontaneous fluctuations in the signal, when there are no specific cognitive demands for the subject (so-called resting state scans). Using spontaneous fluctuations allows us to investigate similarity between regions when it is not biased by any specific task. Therefore, resting-state fMRI can be used to inform us about the inherent organization and functioning of the brain. Gaining a better understanding of the brain's intrinsic architecture and the level of communication this supports is an important basic neuroscience

aim in its own right, and may help us understand how the brain enables complex information processing and rich sets of behaviors, thoughts, and motivations.

Likewise, understanding communication in the brain may also help us understand how things go wrong in a variety of different disorders and neurodegenerative diseases. For example, disorders like attention deficit hyperactivity disorder (ADHD) are thought to be associated with aberrant communication between different regions of the brain (FUENTE *et al.*, 2013). Neurodegenerative diseases such as Alzheimer's and Parkinson's disease are also associated with disruptions in functional connectivity (BASSETT; BULLMORE, 2009).

In addition to the improved understanding of the basic neuroscience of the brain, resting-state fMRI also has great potential to serve as a biomarker for mental disorders (HORWITZ; ROWE, 2011). A biomarker is something that can be measured accurately and reproducibly, and can therefore serve as an objective indication of the medical state of a person. Biomarkers can be used: (i) as indicators of normal biological processes, (ii) for early detection of mental disorders, (iii) as markers of disease progression, (iv) as markers of response to treatment, and (v) for creating personalized treatment strategies that are optimized for the patient.

Resting-state fMRI (rs-fMRI) therefore provides a powerful, non-invasive and interpretable approach to mapping intrinsic brain activity and connectivity. It is particularly advantageous in the study of neurodegenerative diseases, such as Parkinson's disease (PD), where motor impairments can make task-based paradigms difficult or unreliable. By capturing spontaneous synchronization among neural regions, rs-fMRI enables the detection of network-level alterations even in early disease stages. Understanding these intrinsic patterns of communication offers valuable insight into how neurodegeneration disrupts brain organization and function.

The statistical relationships obtained from rs-fMRI data can be represented as networks, where brain regions correspond to nodes and their functional connections to edges. This graph-based representation allows the use of complex network theory to quantify and compare patterns of brain organization between healthy controls and patients with Parkinson's disease.

This study builds upon these insights by leveraging rs-fMRI data to construct and analyze functional brain networks through complex network theory and machine learning, with the goal of developing an automatic and interpretable diagnostic framework for Parkinson's disease.

2.2 Complex Networks and Brain Connectomics

Complex Networks refer to systems with intricate and interconnected structures, characterized by non-trivial patterns of connections among their elements. These networks are often represented as graphs, consisting of nodes (or vertices) and edges (or links) that define relationships between the nodes. This framework has been applied to systems as varied as metabolic networks, food webs, gene–gene interactions, social networks and more recently the human

brain.

Brain connectomics, also referred as network neuroscience, is an approach to understand the brain by recording, analysing and modelling the interactions between its component parts. It is founded upon the idea that the brain comprises a complex web of distinct neural elements. There are two main types of networks in the brain: structural and functional. Structural networks represent the physical connections between brain regions, such as white matter tracts, while functional networks represent statistical dependencies between the activity of different brain regions, often derived from fMRI data.

A first step in understanding the brain network is to characterize what is known about the structure of the brain and its organizing principles. The brain is a complex temporally and spatially multiscale structure that gives rise to elaborate molecular, cellular, and neuronal phenomena (BULLMORE *et al.*, 2009) that together form the physical and biological basis of cognition. The structure within any given scale of the brain is organized into modules, elements that are more highly connected to elements within the same module than to elements outside the module. This modular organization is thought to support a high degree of functional specificity (BASSETT; GAZZANIGA, 2011), robustness to the network and the facility of behavioral adaptation (KIRSCHNER; GERHART, 1998) because each module can both function and change its function without adversely perturbing the remainder of the system.

The physical wiring of these modular elements, the structural network, is far from regular, nor it is random. The modularity itself reveals community structures that segregates the brain into subnetworks with specific cognitive functions. Large-scale brain networks, in contrast, exhibit average path lengths between all pairs of nodes that are much shorter than those of a typical random network (LIAO; VASILAKOS; HE, 2017). This combination of high clustering and short path lengths is characteristic of small-world networks (WATTS; STROGATZ, 1998), which are thought to facilitate efficient information processing by balancing local specialization and global integration.

Apart from small-worldness and modularity, brain networks also feature high-degree nodes, or hubs, acting as bridges between distinct communities, minimizing the overall path lengths across the network and facilitating the integration of information (HEUVEL; SPORNS, 2013).

Although the modular, small-world, and hub structure provide the principles for the organization of brain networks, the overall structure of the brain is determined by a continuous and dynamical weighting of these principles against one another. The structure also considers the metabolic penalty for the formation of long-distance connections and the necessity of short paths between regions with similar functions (VéRTES *et al.*, 2012).

The functional brain network, in contrast, captures the statistical relationships among neural elements that co-vary in activity over time. Nodes represent physical neural elements,

ranging in size from individual neurons to distinct brain regions and edges are defined based on the similarity between two elements' dynamics. The organizing principles of functional brain networks, elucidated using techniques from network science from fMRI data, mirror those of structural networks, exhibiting modularity, small-worldness, and hub architecture (HEUVEL; POL, 2010).

The existence of strong functional community structure, for example, further supports the hypothesis that brain networks segregate into subnetworks with specialized cognitive functions. Moreover, the presence of high clustering and short average path length, combined with the existence of high-degree hub regions, highlights the competing functional pressures of information segregation and integration in the brain (ACHARD *et al.*, 2006).

These network properties can be quantified and compared across populations, offering a powerful framework for investigating how neurological conditions such as Parkinson's disease alter the architecture of functional and structural brain networks.

2.3 Brain Atlases and Regions of Interest (ROIs)

To analyze functional connectivity, the brain must first be divided into a set of predefined regions of interest (ROIs) that serve as the nodes of the network. This spatial division is achieved using a Brain Atlas — a standardized parcellation scheme that labels groups of voxels sharing anatomical or functional characteristics. Atlases provide a consistent framework across subjects, allowing each brain region to be represented by a single BOLD time series obtained by averaging the signal within its voxels.

Brain Atlases can be broadly categorized as anatomical or functional. Anatomical atlases, such as the famous and widely used Automated Anatomical Labeling (AAL) and Harvard–Oxford (HO) atlases, rely on structural landmarks to define cortical and subcortical regions. Functional atlases, on the other hand, identify regions that exhibit similar patterns of activity or connectivity during rest, such as the Shen atlas (SHEN *et al.*, 2013). The choice of atlas determines the resolution and granularity of the resulting network, influencing the computation of graph measures and the interpretability of results.

In this study, AAL3 atlas with 166 ROIS, HO atlas with 132 regions and Shen atlas with 268 ROIS were employed to extract time series from resting-state fMRI data. This parcellation offers a balanced trade-off between spatial detail and computational efficiency, providing a biologically meaningful and reproducible basis for constructing functional connectivity matrices and analyzing the topological organization of the brain in Parkinson's disease.

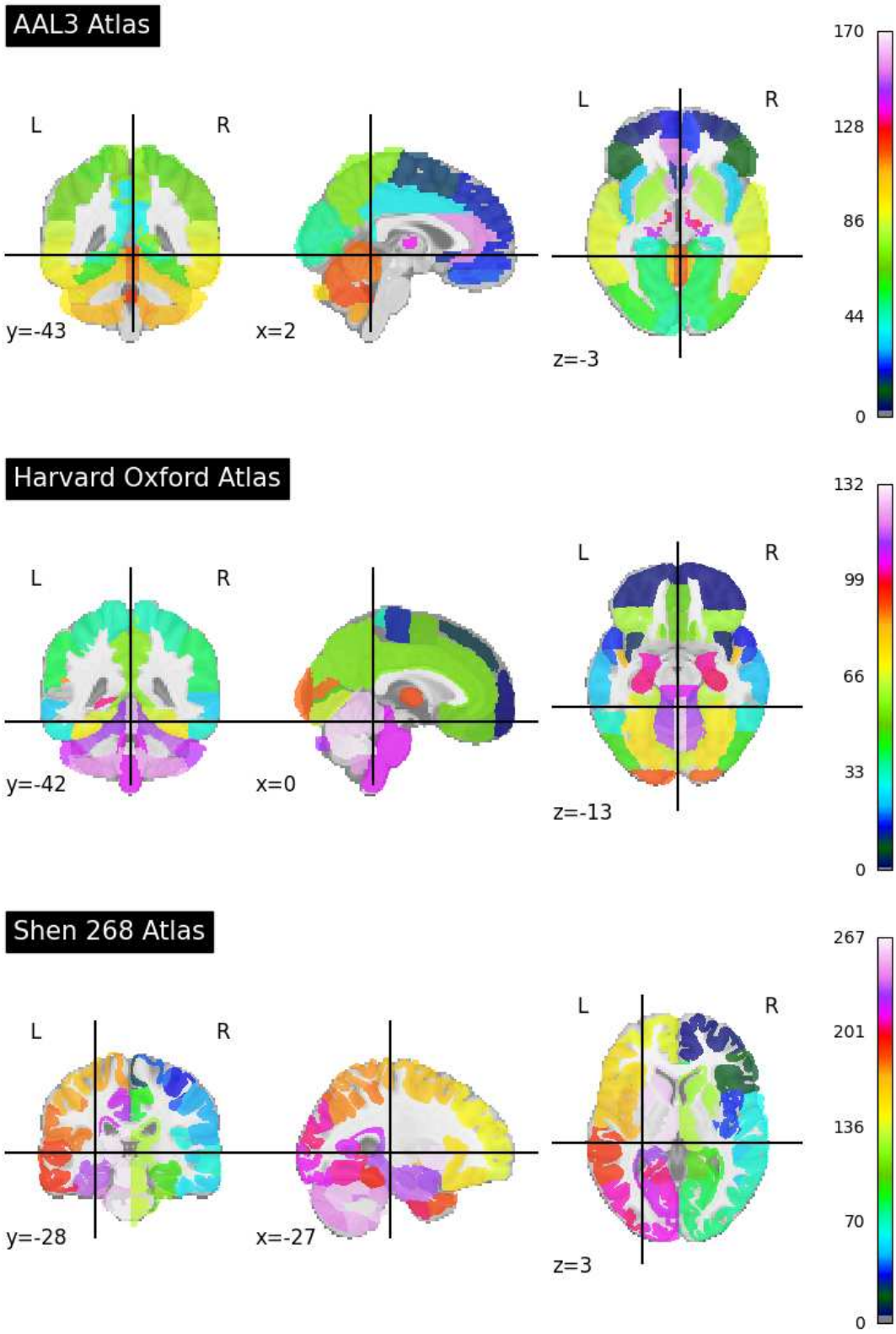

Figure 1 – Partitions of the brain in ROIs for each atlas

Chapter 3

METHODOLOGY

3.1 Overview of the Methodology

This chapter describes the methodology adopted to automatically diagnose Parkinson's disease (PD) from resting-state fMRI data. Figure 2 presents the pipeline, which comprises: (i) data preprocessing, (ii) construction of functional connectivity matrices, (iii) brain network construction and graph-theoretic measures, (iv) ROI-level functional connectivity analysis, and (v) machine-learning classification with medical interpretation. Each step was designed to ensure reproducibility, interpretability, and robustness.

3.2 Data Acquisition

The dataset used in this study was obtained from the Parkinson's Progression Markers Initiative (PPMI). PPMI is a comprehensive, multi-center study that collects clinical, imaging, and biological data from individuals with Parkinson's disease (PD) and healthy controls. The dataset includes resting-state fMRI and anatomical MRI scans for three cohorts: Healthy Controls (HC), early-stage PD patients (Prodromal), and late-stage PD patients (PD).

According to the PPMI Study Cohort documentation, Prodromal patients are those participants who are at risk of Parkinson's based on clinical features, genetic variants, or other biomarkers, but who have not been diagnosed. PD consists of patients who have a recent formal diagnosis of Parkinson's disease that were untreated and have pathogenic genetic variant(s). Finally, Controls are those participants with no neurologic disorder and no first-degree relative with PD.

Table 1 – Demographic information (PPMI subjects included prior to QC).

	Prodromal (n = 249)	**PD (n = 185)**	**Controls (n = 74)**
Age (years)	66.00 ± 4.78	64.05 ± 6.68	65.80 ± 10.56
Sex (M/F)	143/106	127/58	41/33

The data acquisition protocols were standardized across all participating sites to ensure consistency and reliability. All MRI images were extracted from MRI machines with a magnetic field strength of 3 Tesla, slice thickness of 3.5 mm, Z matrix of 256, acquisition plane SAGITTAL, weighting T1, and machines from the following manufacturers: Philips, GE, Siemens. In addition,

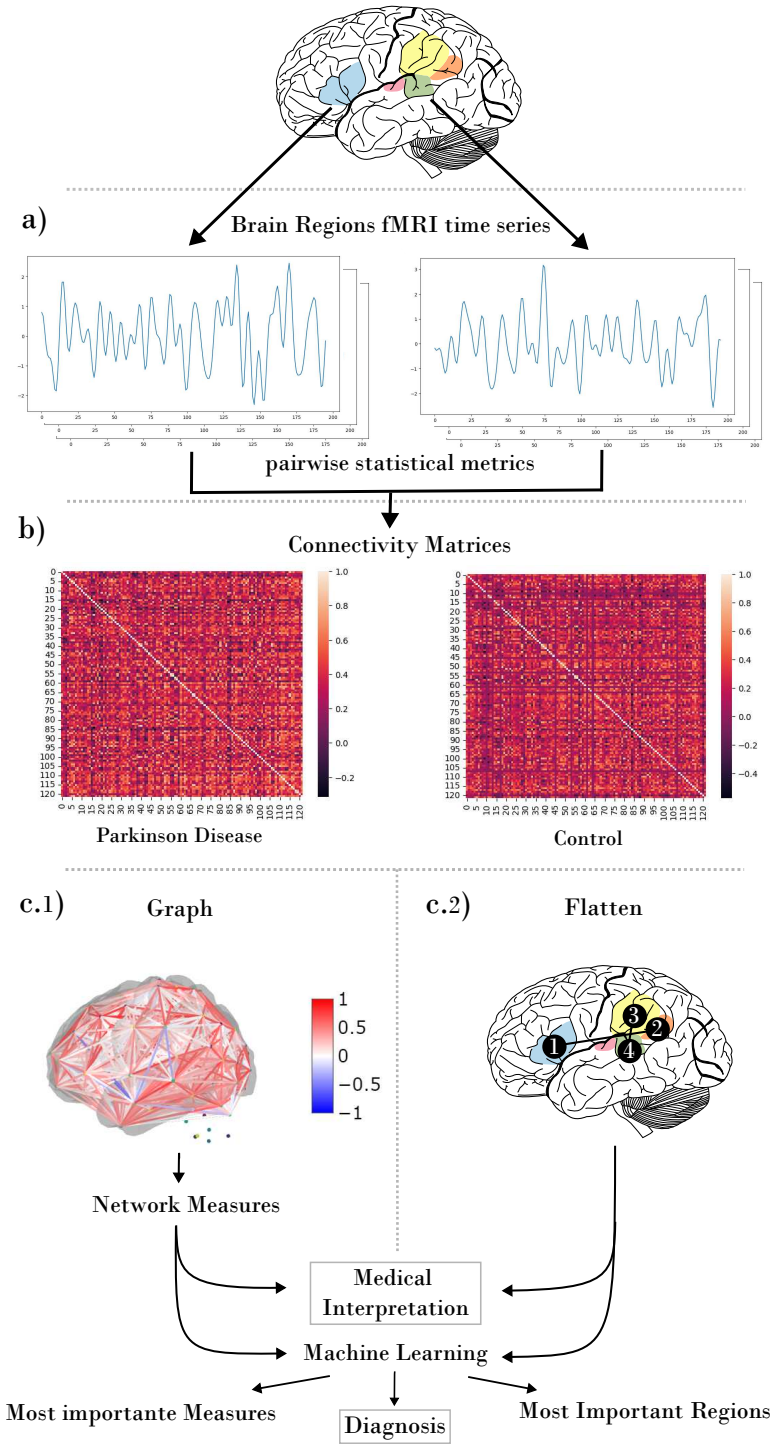

Figure 2 – Methodology Overview

all selected fMRI images were extracted from machines with a magnetic field strength of 3 Tesla, slice thickness of 3.5 mm, echo time (TE) of 30 ms, repetition time (TR) of 2500 ms, flip angle between 80º and 90º, and machines from the following manufacturers: Philips, GE, Siemens.

3.3 Data Preprocessing

The preprocessing of resting-state fMRI data is a critical step to ensure the quality and reliability of subsequent analyses. Raw fMRI data often contain various sources of noise and artifacts that can affect the interpretation of functional connectivity patterns. There are two primary sources of structured noise in resting state fMRI acquisitions, namely the physiology (i.e., breathing and heartbeat), movement of the subject while they are in the scanner and scanner-induced noise.

In order to mitigate these previous issues, we employed a comprehensive preprocessing pipeline using the CONN toolbox (NIETO-CASTANON, 2020), which is built on top of SPM (Statistical Parametric Mapping) software.

The main preprocessing steps include:

3.3.1 Phase Encoding Bias Correction

Among the raw data provided by PPMI are, for each patient, fMRI data in Left-to-Right (LR) and Right-to-Left (RL) phase encodings. LR and RL acquisitions have opposite phase encoding directions, which leads to spatial distortions and compressions of the image in opposite directions. This causes geometric distortion and signal loss in different areas of the brain for each run. If not corrected, differential distortions can introduce asymmetries in connectivity patterns that are unrelated to neural activity, thus compromising the anatomical validity and statistical comparability of the results.

To eliminate this acquisition bias and improve the spatial accuracy of the data, the topup tool from the FSL (FMRIB Software Library) software was used. The procedure consisted of extracting a reference image (SBRef) from each acquisition — one with LR encoding and one with RL encoding. The two images were combined into a single 4D volume using the fslmerge command, and a parameter file (acqparams.txt) was created containing information about the phase encoding direction and total readout time for each acquisition.

Next, the topup command was used to estimate a distortion map based on the differences between the two images. This map represents the displacement field caused by inhomogeneities in the magnetic field. After the estimation, the applytopup command was used to apply the correction to the entire time series of rs-fMRI data in both directions (LR and RL), resulting in corrected and aligned images.

After this process, 8 images from the Control class, 12 PDs, and 19 Prodromal images were excluded due to missing metadata or correction failures.

3.3.2 Motion Correction

The first step in this pre-processing, which is commonly applied to all fMRI data, is motion correction, with the aim of correcting for the effect of the subject's head movement in the scanner. Functional data are realigned using the SPM12 realign & unwarp tool procedure based on Andersson *et al.* (2001). Motion correction applies transformations to each volume so that all resulting volumes are combined and spatially aligned in time. In addition to aligning volumes, motion correction also provides results from head motion estimates throughout the scan, such as a set of motion parameters. These motion parameters can be used to correct for motion in subsequent analyses.

3.3.3 Slice-Timing Correction

Instead of collecting data spatially from the entire brain at each moment in time, each fMRI machine collects partial 'slices' of the brain, one at a time, until the whole is collected. The time it takes to collect all the partial slices of the brain until the whole is complete is called TR (Repetition Time). The purpose is to correct for the small difference in the time each slice of BOLD data is acquired (i.e., some slices are acquired at the beginning of the TR (Repetition Time), while others are acquired at the end of the TR). When the TR of a sequence is, for example, 3 seconds,the difference in acquisition time for each slice can vary greatly, and it may be useful to apply slice timing correction. Slice timing correction uses time interpolation to slightly shift the BOLD time courses of the voxels to account for these small differences in acquisition time. The correction is performed using the Slice-Timing Correction (STC) procedure of SPM12 based on Henson *et al.* (1999). As all of our data has a TR of 2.5 seconds, we applied this step to all images.

3.3.4 Outlier Identification

Scans with possible outliers are identified based on the global BOLD signal observed and the amount of movement of the subject in the scanner. Acquisitions with frame displacement above 0.9 mm or global changes in the BOLD signal above 5 standard deviations were flagged as possible outliers. Frame displacement is calculated at each time point, considering a bounding box of 140x180x115 mm around the brain and estimating the largest displacement between six control points placed at the centre of the faces of this bounding box. The change in global BOLD signal is calculated at each time point as the change in the average BOLD signal within the SPM global average mask, scaled to standard deviation units. At this stage, 2 invalid scans were identified in the Control class, 4 invalid scans in the PD class, and 7 invalid scans in the Prodromal class.

3.3.5 Direct Segmentation and Normalization

Functional (fMRI) and anatomical (MRI) data are normalised in MNI space, defined as a standard brain using a large series of MRI scans in normal controls, and segmented into grey matter, white matter, and CSF (Cerebrospinal Fluid) using the unified segmentation and normalisation procedure of SPM12. This procedure iteratively performs tissue classification, estimating posterior tissue probability maps (TPMs) from the intensity values of the reference functional/anatomical image, and registration, estimating the non-linear spatial transformation that best approximates the posterior and anterior TPMs, until convergence. Direct normalisation applies this unified segmentation and normalisation procedure separately to functional data, using the average BOLD signal as a reference image, and to anatomical data.

3.3.6 Spatial Smoothing

An additional step that is commonly applied to various types of fMRI data in the preprocessing pipeline is the application of spatial smoothing to the image. Spatial smoothing is achieved by calculating, in each voxel, a weighted average of several neighbouring voxels and has the effect of blurring the images. The amount of spatial smoothing applied was set at 8 mm (the standard value used) of the full-width-half-maximum (FWHM) of the Gaussian kernel used to create the weights for the average. The advantage of spatial smoothing is that calculating the average helps to reduce the influence of noise. However, the disadvantage of spatial smoothing is that some of the spatial location accuracy is lost. Figure 3 illustrates this spatial smoothing step.

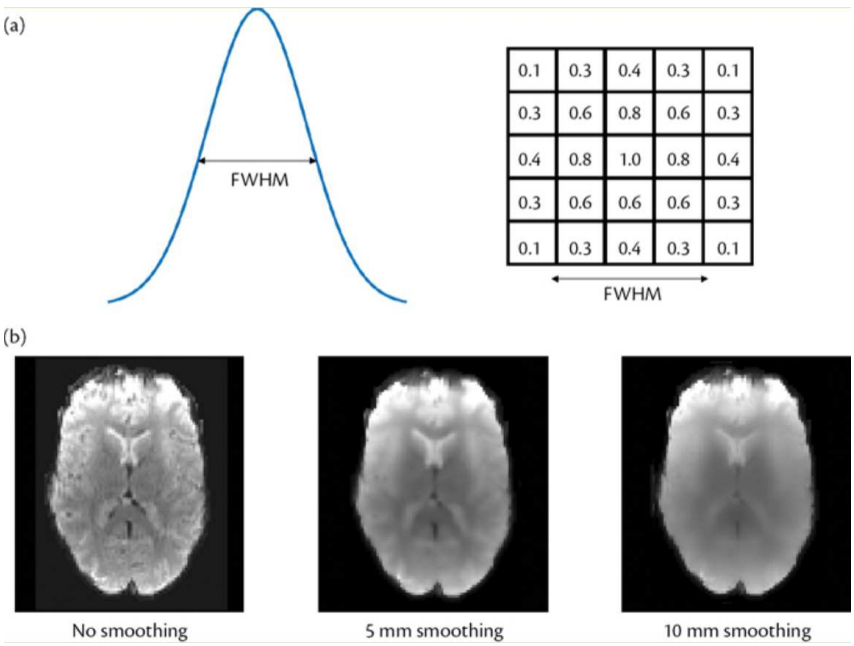

Figure 3 – Illustration from the book An Introduction to Resting State fMRI Functional Connectivity (BIJSTER-BOSCH; SMITH; BECKMANN, 2017)

3.3.7 Band-pass temporal filtering

Temporal frequencies below 0.008 Hz or above 0.09 Hz are removed from the BOLD signal in order to focus on slow-frequency fluctuations while minimizing the influence of physiological, head-motion and other noise sources. Filtering is implemented using a discrete cosine transform windowing operation to minimize border effects, and performed after regression to avoid any frequency mismatch in the nuisance regression procedure (HALLQUIST; HWANG; LUNA, 2013).

3.3.8 Voxel clustering into Regions of Interest (ROIs)

To reduce the dimensionality of the data and facilitate the analysis of functional connectivity, the preprocessed fMRI data were parcellated into Regions of Interest (ROIs) across three different atlases: AAL3, Harvard-Oxford and Shen. For each ROI, the mean BOLD signal time series was extracted by averaging the signals of all voxels within that region. This approach not only reduces computational complexity but also enhances the interpretability of the results by focusing on functionally relevant brain regions.

These atlases were selected to test the robustness of our findings across different parcellation schemes: AAL3 is a widely-used anatomical atlas, Harvard-Oxford provides probabilistic anatomical regions, and the Shen atlas is a functionally-defined parcellation. Comparing results across them will allow us to see if our findings are dependent on a specific atlas definition.

3.4 Construction of Functional Connectivity Matrices

Once all BOLD time series from the three atlases ROIs were extracted, one way to address how connectivity, integration of information and communication is structured between brain regions, is to adopt a node-based connectivity approach based on graph's 'nodes and edges'. Modelling this way, each ROI represents a node in the brain network, and the functional connections between these nodes are represented as edges. The strength of these connections between pairs of ROIs BOLD timeseries is typically quantified using statistical measures such as Pearson's correlation coefficient, Spearman correlation, Partial Correlation, Granger causality, Mutual Information (MI). However, in this work, we adopted the Pearson's correlation coefficient due to its widespread use and effectiveness (LUPPI *et al.*, 2024; LIANG *et al.*, 2012). After computing pairwise correlations across all ROIs, we obtained a symmetric connectivity matrix A for each subject, where the elements A_{ij} and A_{ji} correspond to the correlation between the pair of regions i and j.

3.5 Brain Network Analysis

Graph Theory and Network Science measures provide valuable insights into the brain's topology and organization. The Connectivity Matrix 3.4 will be used in this section as if it was a graph adjacency matrix, where each ROI represents a node and the correlation values between them represent the edges.

For this study, we considered three methods to build the brain network from the connectivity matrix: (i) Weighted Undirected Network with fixed threshold, and (ii) Weighted Undirected Network with fixed density and (iii) Weighted Undirected Network via Orthogonal Minimal Spanning Tree algorithm. These three methods were chosen to systematically test different network construction assumptions. Method (i) represents a common, straightforward thresholding approach, while method (ii) provides a more robust comparison by ensuring all networks have the same edge density. Finally, method (iii) utilizes a modern algorithm to create a sparse but fully connected network, allowing us to investigate network properties that are not dependent on a specific threshold or density hyperparameter.

After the construction of the brain networks for each subject, we extracted a set of complex network measures that capture various aspects of the network's topology and organization. These measures include: number of nodes, number of edges, density, average clustering coefficient, average shortest path length, global efficiency, local efficiency, assortativity, modularity, transitivity, average betweenness centrality, average closeness centrality, average eigenvector centrality, degree variance, radius and diameter (COSTA *et al.*, 2007). Each of these measures provides insights into different facets of brain network organization, such as integration, segregation, resilience, and centrality of nodes within the network.

These measures will be used as features in machine learning models to classify subjects into different groups (Healthy Controls, prodromal, and PD) and to investigate potential alterations, biomarkers and medical interpretations about brain network topology associated with Parkinson's disease progression.

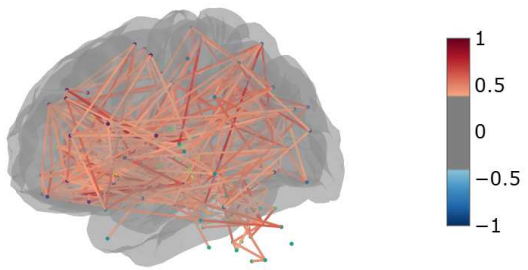

Figure 4 – Example of a Weighted Undirected Brain Network with fixed threshold

3.6 Regions of Interest (ROIs) Functional Connectivity Analysis

In addition to the brain network analysis, we also performed a Region of Interest (ROI) analysis to investigate specific brain regions functional connections that may be implicated in Parkinson's disease (PD). While network metrics capture global topological changes, ROI-level analysis allows the identification of localized connectivity alterations in regions known to be affected by PD, such as the basal ganglia and motor cortex. This analysis focuses on examining the FC patterns and alterations within predefined ROIs known to be associated with PD pathology.

Some of the questions we aim to address with this analysis include: Which brain regions consistently show altered connectivity patterns in PD patients compared to healthy controls? Are there specific ROIs whose connectivity strength correlates with disease severity or progression?

For this purpose, each subject connectivity matrix is used as features for the classification of PD patients and Healthy Controls and for further analysis.

3.7 Machine Learning Classification

The final stage of our methodology involves applying machine learning techniques to classify subjects based on the extracted features from both brain network analysis and ROI functional connectivity analysis. We will explore various supervised learning algorithms, such as Logistic Regression, Support Vector Machines (SVM), Random Forests, and Gradient Boosting and Naive Bayes to build robust classification models.

To assess model performance reliably, especially under class imbalance and under the multi-class classification condition, we used the F1-score as the primary metric. F1-score is the harmonic mean of precision and recall, providing a balanced measure that accounts for both false positives and false negatives, essential for medical classification tasks where the cost of misclassification is high.

In addition, we employed techniques such as Synthetic Minority Over-sampling Technique (SMOTE), class weighting to address class imbalance during model training and stratified k-fold cross-validation to ensure robust evaluation of model performance. Hyperparameter tuning will be conducted using grid search or random search methods to optimize model parameters for improved classification accuracy.

3.8 Medical Interpretation

A central goal of this research is not only to classify patients but also to transform the model's findings into interpretable biological insights. This final step is essential for validating the classifier, building clinical trust, and elucidating the neural mechanisms of Parkinson's disease. To achieve this, we will employ a three-pronged approach to analyze the features from both the network topology and the ROI-to-ROI connectivity.

To facilitate this interpretation, the 268 individual ROIs from the Shen atlas are grouped into 8 established resting-state functional networks: Medial Frontal, Frontoparietal, Default Mode, Subcortical-Cerebellum, Motor, Visual I, Visual II, and Visual association (SHEN *et al.*, 2013). This aggregation allows us to move from individual connection-level changes to a more comprehensive, large-scale network perspective.

- **Group-Level Connectivity Visualization:** To identify which functional connections are most consistently altered by the disease, we will perform a group-level statistical comparison. This involves calculating the median connectivity matrix for each class (Control, Prodromal, PD) and then computing difference maps (e.g., median squared difference) between the patient groups and the healthy controls. These differences will be visualized using connectome circle plots and network-level heatmaps to identify the specific connections and large-scale networks most impacted by the disease.
- **Brain Network Topology Analysis:** To assess how PD affects the global and local organization of the brain, the distributions of the graph-theoretic measures (from Section 3.5) will be statistically compared across the three groups. This analysis will be visualized using boxplots to identify any consistent trends (e.g., increases or decreases) in metrics like local efficiency, modularity, or assortativity that correlate with disease progression from healthy controls to prodromal and PD patients.
- **Machine Learning Explainability (SHAP):** To understand which features the classification models are using to make their decisions, we will apply SHAP (SHapley Additive exPlanations). A Random Forest classifier will be trained on the full set of pairwise ROI-to-ROI connectivity features. We will then compute global SHAP values, which quantify the contribution of each individual connection to the model's overall predictive power. These SHAP values will be aggregated by their respective brain networks and visualized in a heatmap to reveal the most discriminative connections and networks for the classifier.

Finally, the results from these three analytical methods will be synthesized and interpreted in the context of existing neuroscientific literature on PD. This allows us to validate that our model is learning biologically plausible patterns and to contribute to the broader understanding of PD as a network-level brain disorder.

Chapter 4

RESULTS

4.1 Introduction

This chapter presents the results of the classification pipeline. First, we present the comparative performance of the two primary feature sets: (1) graph-based network measures and (2) ROI-based functional connectivity. We evaluate these features across all multi-class classification tasks and all atlases. Second, we identify the single best-performing model and feature set. Finally, we analyse this model to identify the most discriminative biomarkers for Parkinson's disease.

The Comparative Classification results are reported as mean $\pm$ standard deviation across the outer cross-validation folds. We report macro F1 as the primary metric due to class imbalance in the three-class setting (Control, Prodromal, Parkinson), and include accuracy and weighted F1 for completeness. The baseline model was built according to Zero-Rule (ZeroR) or majority class classifier. All model selection and hyperparameter tuning were nested within cross-validation to prevent leakage. Class rebalancing (SMOTE) and PCA, when used, were fit inside the training folds only.

4.2 Comparative Classification Performance

4.2.1 Approach 1: Classification using Graph-Based Network Measures

In the first approach, we extracted 17 network-level topological measures as mentioned in 3.5 (e.g., clustering coefficient, global efficiency, modularity) from the connectivity matrices. These measures were used as features to train five different classifiers: Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), XGBoost, and Naive Bayes (NB). The experiment was repeated for all combinations of the three brain atlases and three graph construction methods.

The results for the multi-class classification task are summarized in Table 2.

A detailed analysis of Table 2 reveals several critical patterns. The overall classification performance for this feature set is modest. The top-performing pipeline—Logistic Regression

Table 2 – Multi-class performance using graph-based network measures across atlases, graph construction strategies, and classifiers. Values are mean ± std across folds.

Atlas	Construction	Classifier	Test Acc. ± std	Test F1 (w) ± std	Test F1 (m) ± std	Train Acc.	Train F1 (w)	Train F1 (m)
		Baseline	0.463 ± 0.000	0.293 ± 0.000	0.211 ± 0.000	—	—	—
AAL3	Absolute Threshold	LR	0.350 ± 0.041	0.366 ± 0.042	0.339 ± 0.041	0.457	0.457	0.456
		RF	0.325 ± 0.042	0.322 ± 0.045	0.253 ± 0.048	1.000	1.000	1.000
		SVM	0.325 ± 0.054	0.335 ± 0.060	0.300 ± 0.062	0.902	0.902	0.900
		XGBoost	0.338 ± 0.019	0.346 ± 0.017	0.272 ± 0.020	0.962	0.962	0.961
		NB	0.300 ± 0.034	0.298 ± 0.036	0.284 ± 0.034	0.376	0.366	0.370
	Fixed Density	LR	0.237 ± 0.028	0.238 ± 0.026	0.229 ± 0.024	0.476	0.472	0.466
		RF	0.325 ± 0.028	0.332 ± 0.032	0.273 ± 0.033	1.000	1.000	1.000
		SVM	0.325 ± 0.042	0.333 ± 0.046	0.260 ± 0.047	0.758	0.756	0.751
		XGBoost	0.275 ± 0.080	0.291 ± 0.074	0.229 ± 0.074	0.938	0.938	0.939
		NB	0.237 ± 0.075	0.239 ± 0.083	0.215 ± 0.080	0.428	0.413	0.410
	OMST	LR	0.375 ± 0.045	0.385 ± 0.047	0.354 ± 0.046	0.450	0.445	0.442
		RF	0.287 ± 0.032	0.292 ± 0.037	0.250 ± 0.037	1.000	1.000	1.000
		SVM	0.400 ± 0.023	0.411 ± 0.032	0.389 ± 0.034	0.739	0.735	0.732
		XGBoost	0.375 ± 0.018	0.366 ± 0.017	0.338 ± 0.017	0.789	0.788	0.785
		NB	0.325 ± 0.049	0.345 ± 0.042	0.297 ± 0.041	0.402	0.391	0.391
Harvard-Oxford	Absolute Threshold	LR	0.275 ± 0.031	0.272 ± 0.029	0.263 ± 0.029	0.507	0.502	0.493
		RF	0.375 ± 0.058	0.377 ± 0.065	0.336 ± 0.066	1.000	1.000	1.000
		SVM	0.300 ± 0.033	0.313 ± 0.032	0.270 ± 0.029	0.873	0.872	0.870
		XGBoost	0.400 ± 0.047	0.396 ± 0.045	0.366 ± 0.048	0.981	0.981	0.980
		NB	0.300 ± 0.050	0.303 ± 0.048	0.289 ± 0.047	0.409	0.379	0.368
	Fixed Density	LR	0.388 ± 0.017	0.398 ± 0.022	0.372 ± 0.023	0.500	0.500	0.496
		RF	0.412 ± 0.036	0.419 ± 0.037	0.366 ± 0.039	1.000	1.000	1.000
		SVM	0.338 ± 0.044	0.355 ± 0.046	0.322 ± 0.046	0.797	0.794	0.791
		XGBoost	0.362 ± 0.042	0.360 ± 0.039	0.318 ± 0.039	0.811	0.809	0.805
		NB	0.263 ± 0.034	0.230 ± 0.064	0.227 ± 0.057	0.378	0.348	0.334
	OMST	LR	0.400 ± 0.027	0.421 ± 0.025	0.367 ± 0.027	0.512	0.511	0.506
		RF	0.425 ± 0.055	0.419 ± 0.055	0.376 ± 0.051	1.000	1.000	1.000
		SVM	0.350 ± 0.052	0.360 ± 0.050	0.322 ± 0.050	0.778	0.776	0.773
		XGBoost	0.400 ± 0.039	0.407 ± 0.042	0.388 ± 0.041	0.873	0.872	0.869
		NB	0.200 ± 0.030	0.184 ± 0.041	0.189 ± 0.046	0.433	0.416	0.406
Shen 268	Absolute Threshold	LR	0.425 ± 0.022	0.432 ± 0.022	0.424 ± 0.021	0.557	0.553	0.546
		RF	0.412 ± 0.040	0.418 ± 0.037	0.394 ± 0.035	1.000	1.000	1.000
		SVM	0.412 ± 0.017	0.412 ± 0.017	0.378 ± 0.015	0.835	0.833	0.829
		XGBoost	0.475 ± 0.042	0.477 ± 0.035	0.456 ± 0.034	0.964	0.964	0.964
		NB	0.362 ± 0.032	0.340 ± 0.033	0.341 ± 0.032	0.455	0.434	0.435
	Fixed Density	LR	0.487 ± 0.038	0.503 ± 0.039	0.478 ± 0.043	0.517	0.516	0.511
		RF	0.375 ± 0.070	0.372 ± 0.070	0.334 ± 0.069	1.000	1.000	1.000
		SVM	0.400 ± 0.044	0.407 ± 0.042	0.383 ± 0.041	0.720	0.719	0.714
		XGBoost	0.300 ± 0.036	0.311 ± 0.032	0.283 ± 0.031	0.923	0.923	0.922
		NB	0.175 ± 0.039	0.089 ± 0.060	0.123 ± 0.065	0.383	0.283	0.271
	OMST	LR	0.338 ± 0.030	0.348 ± 0.029	0.337 ± 0.032	0.548	0.549	0.545
		RF	0.400 ± 0.039	0.369 ± 0.034	0.305 ± 0.033	1.000	1.000	1.000
		SVM	0.388 ± 0.052	0.386 ± 0.043	0.360 ± 0.044	0.806	0.804	0.801
		XGBoost	0.475 ± 0.035	0.457 ± 0.036	0.391 ± 0.038	0.950	0.950	0.949
		NB	0.312 ± 0.038	0.309 ± 0.035	0.308 ± 0.033	0.500	0.497	0.494

(LR) with the Shen atlas and Fixed Density construction—achieved a mean test macro F1-score (Test F1 (m)) of 0.478. Although superior to the 0.211 F1-score of the Baseline model, it indicates that the extracted graph-based measures still possess limited discriminatory power. The analysis suggests this limitation stems not from a specific classifier, but from a fundamental lack of signal in the graph-theoretic features themselves.

The most striking trend across all 45 experimental permutations is the profound evidence of model overfitting, particularly in high-capacity, non-linear classifiers like RF. This overfitting pattern is corroborated by the Support Vector Machine (SVM) and XGBoost models. These classifiers also demonstrate a dramatic disconnect between high training performance (e.g., 0.902 for AAL3/SVM; 0.981 for H-O/XGBoost) and low test performance (0.325 and 0.400, respectively). In contrast, the low-capacity linear models, Logistic Regression (LR) and Naive Bayes (NB), show much closer training and test scores. Their low training accuracies (e.g., LR:

0.457, NB: 0.376 for AAL3/Absolute Threshold) indicate that they are not overfitting; rather, their performance is limited by the features themselves, which are not linearly separable.

To clarify the impact of atlas choice, the best-performing model for each atlas is summarized in Table 3.

Table 3 – Best-performing model (Test F1-macro) for each Atlas in Approach 1.

Atlas	Best F1-macro ± std	Best Classifier	Best Construction
AAL3	0.389 ± 0.034	SVM	OMST
Harvard-Oxford	0.388 ± 0.041	XGBoost	OMST
Shen 268	0.478 ± 0.043	LR	Fixed Density

This summary table reveals a clear and significant finding: the choice of brain parcellation is a dominant factor in model performance. The Shen 268 atlas (F1-m 0.478) substantially outperforms both the AAL3 (F1-m 0.389) and Harvard-Oxford (F1-m 0.388) atlases, which perform nearly identically. This result aligns with extensive findings in neuroimaging literature (REVELL *et al.*, 2022). The AAL3 and Harvard-Oxford atlases are structural parcellations, which define regions of interest (ROIs) based on anatomical boundaries. A known limitation of this approach is that a single anatomical ROI may contain multiple, functionally distinct signals; averaging the BOLD signal across these heterogeneous voxels can obscure or destroy the very functional information the model seeks to find. The Shen 268 atlas, conversely, is a functionally-derived atlas, created by clustering resting-state fMRI data to produce 268 regions of homogeneous functional connectivity. Studies consistently show that finer-grained, functional atlases yield superior classification performance because their nodes provide a more valid representation of the brain's true functional units (SOHN *et al.*, 2015).

4.2.2 Approach 2: Classification using ROI-Based Functional Connectivity

The results for the multi-class classification task with ROI Functional Connectivity are summarized in Table 4.

This approach, which uses the raw ROI-to-ROI functional connectivity matrix as features, yielded dramatically superior results. The top-performing model, Shen 268 + Logistic Regression, achieved a test F1-macro score of 0.665 ± 0.064. This represents a 39% relative improvement over the best model from Approach 1 (0.478) and strongly supports the hypothesis that the raw connectivity patterns contain more discriminative information than the abstracted graph-based topological measures.

This approach introduces the "curse of dimensionality" ($p \gg n$), where the feature space p (e.g., $p = 35,778$ for Shen 268) vastly exceeds the sample size n. This high dimensionality rendered complex non-linear models (SVM, XGBoost) computationally intractable. It also

Table 4 – ROI-based functional connectivity classification across atlases and classifiers. Values are mean ± std across folds. SVM and XGBoost were discarded in this experiment due to excessive training time.

Atlas	Classifier	Test Acc. ± std	Test F1 (w) ± std	Test F1 (m) ± std	Train Acc.	Train F1 (w)	Train F1 (m)
	Baseline	0.463 ± 0.000	0.293 ± 0.000	0.211 ± 0.000	—	—	—
AAL3	LR	0.550 ± 0.038	0.546 ± 0.035	0.485 ± 0.044	1.000	1.000	1.000
	LR_PCA	0.588 ± 0.036	0.589 ± 0.038	0.560 ± 0.038	0.731	0.734	0.717
	RF	0.438 ± 0.036	0.396 ± 0.051	0.309 ± 0.061	1.000	1.000	1.000
	NB	0.463 ± 0.068	0.448 ± 0.063	0.405 ± 0.048	0.900	0.892	0.856
Harvard-Oxford	LR	0.562 ± 0.030	0.561 ± 0.029	0.493 ± 0.044	1.000	1.000	1.000
	LR_PCA	0.487 ± 0.014	0.500 ± 0.017	0.472 ± 0.028	0.756	0.758	0.740
	RF	0.412 ± 0.065	0.378 ± 0.068	0.296 ± 0.066	1.000	1.000	1.000
	NB	0.463 ± 0.040	0.450 ± 0.038	0.401 ± 0.028	0.863	0.856	0.817
Shen 268	LR	0.688 ± 0.044	0.686 ± 0.050	0.665 ± 0.064	1.000	1.000	1.000
	LR_PCA	0.588 ± 0.051	0.589 ± 0.052	0.597 ± 0.067	0.841	0.841	0.853
	RF	0.588 ± 0.059	0.541 ± 0.067	0.560 ± 0.078	1.000	1.000	1.000
	NB	0.450 ± 0.070	0.426 ± 0.069	0.366 ± 0.062	0.925	0.920	0.892

induced severe overfitting, evidenced by 1.000 training accuracies. In this high-dimensional space, the simpler, linear model (LR) proved more robust (Test Acc. 0.688) than the non-linear RF (Test Acc. 0.588), which was more susceptible to memorizing noise.

The Principal Component Analysis experiment (LR_PCA) provides further nuance. PCA acted as a regularizer, mitigating overfitting and improving performance for the coarse, noisy AAL3 atlas (F1-m 0.560 vs. 0.485). In contrast, this dimensionality reduction degraded performance for the fine-grained Shen 268 atlas (F1-m 0.597 vs. 0.665). This suggests PCA discarded subtle, high-dimensional discriminative information captured by the functional atlas, reinforcing the superiority of the raw, high-dimensional feature set for this parcellation.

Finally, the confusion matrix for the top-performing model, Shen 268 + Logistic Regression:

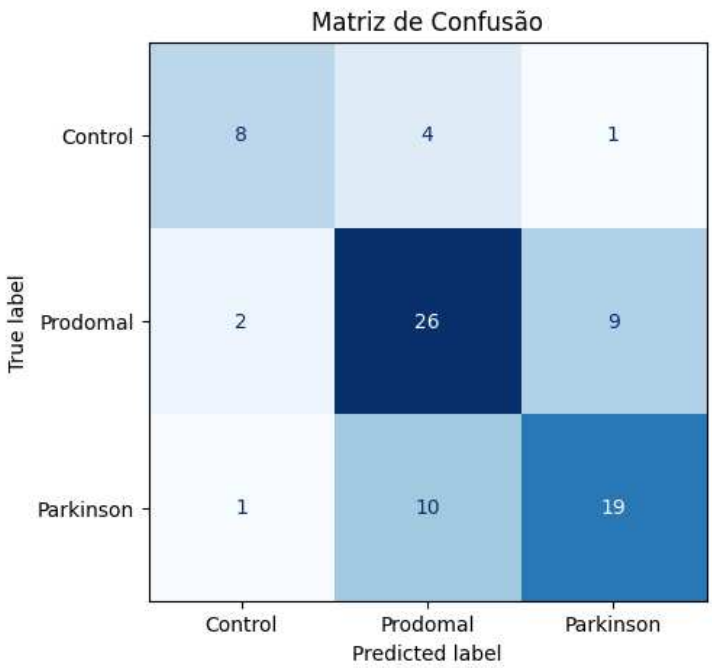

Figure 5 – Test set confusion matrix of the best obtained model

4.3 Medical Interpretation: Identifying Biomarkers

4.3.1 Visualization of Altered Connectivity

To better understand the nature of the connectivity changes identified by the model, this section visualizes the differences between the patient groups.

Figure 6 illustrates the mean connectivity differences between the groups using connectome circle plots. These plots highlight the specific connections that show the most significant changes.

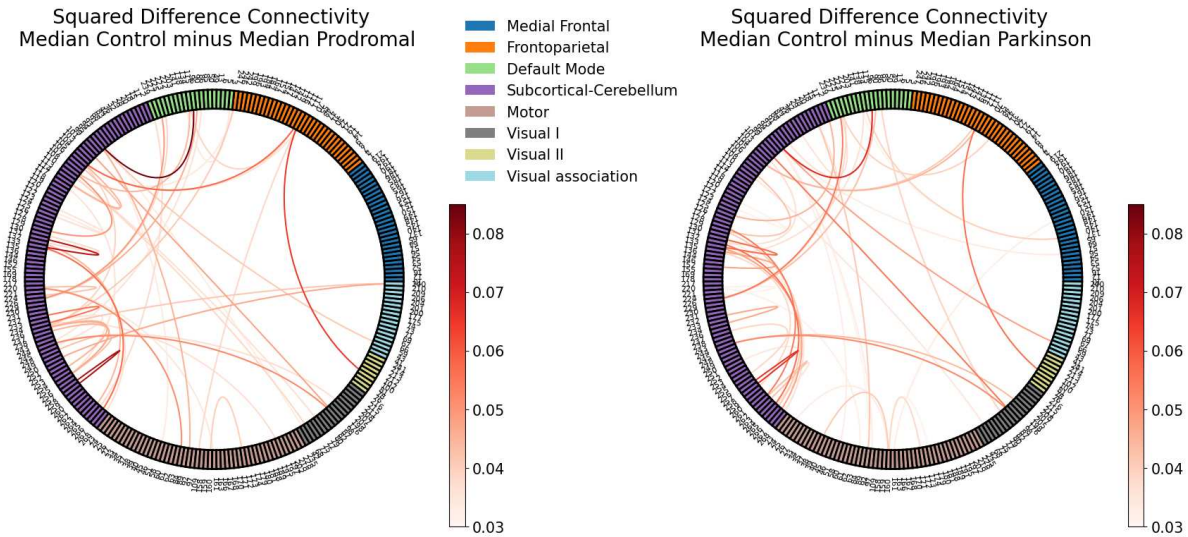

Figure 6 – Connectome visualization of mean connectivity differences. Source: `visualization.ipynb`.

Furthermore, Figure 7 presents a heatmap of the squared difference between the mean connectivity matrices of the control and Parkinson's disease groups. This visualization provides a comprehensive map of altered connectivity, with brighter areas indicating larger differences.

As summarized in Table 6 all of the top 50 functional connections with the largest group differences showed decreased connectivity in the PD group relative to controls. These largest changes were not evenly distributed across the brain, but instead showed a pronounced network-specific pattern in Figure 7. In particular, the subcortical–cerebellar network showed the most extensive connectivity reductions in Prodromal and in PD. Within the SC network itself, 21 of the top 50 altered connections were within SC links. This loss of intra-network connectivity suggests a weakening of functional coupling within subcortical and cerebellar circuits in PD. In practical terms, brain regions comprising the basal ganglia, thalamus, and cerebellum (which are grouped in the SC functional network) were far less synchronized with each other in PD compared to healthy controls. This pattern aligns with emerging views of PD as a network disorder that preferentially disrupts subcortical–cortical communication (TANG *et al.*, 2022).

In addition to these within-SC changes, PD was marked by markedly reduced connectivity between the SC network and multiple cortical networks. Among the top differences, 6 involved

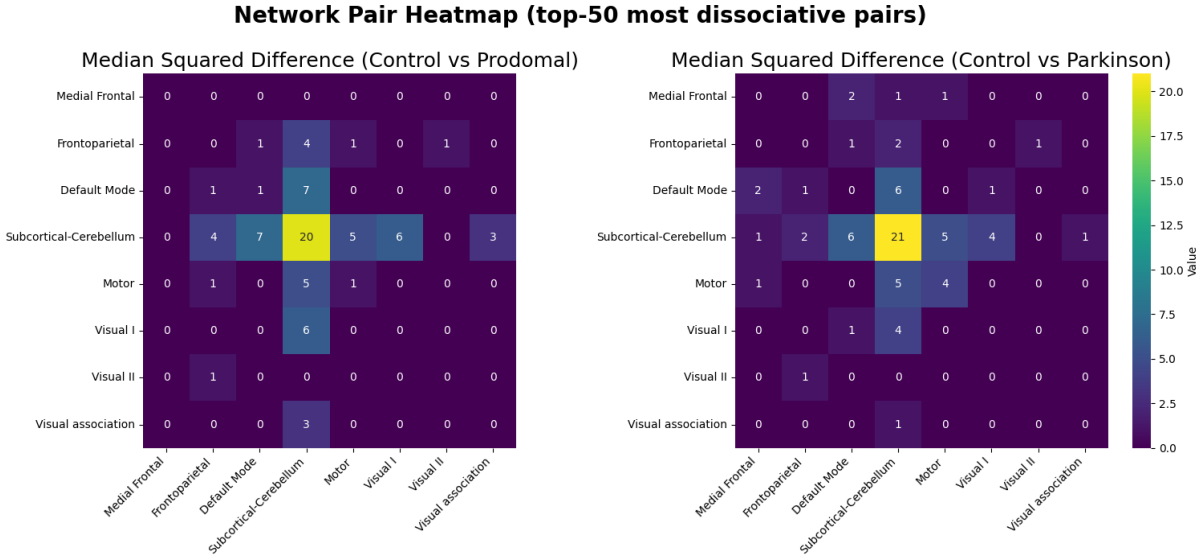

Figure 7 – Heatmap of the squared difference in mean connectivity between Control and PD groups. Source: `visualization.ipynb`.

connections between SC regions and the default mode network, 5 between SC and the motor network, and 5 between SC and visual network regions (combining primary and association visual networks). A smaller number of top connections (2 of 50) linked SC regions to the frontoparietal (executive) network, and one involved an SC link to the medial frontal network. PD appears to isolate the SC network from much of the rest of the brain's functional connectome. This widespread disconnection is consistent with recent reports that PD patients exhibit significant reductions in connectivity both within subcortical–cerebellar circuits and between the cerebellum and multiple large-scale cortical networks (HE *et al.*, 2025). This widespread reorganization is why PD has diverse symptoms beyond motor deficits. Researchers have described PD as a "disconnection syndrome" (GöTTLICH *et al.*, 2013).

Our findings with this analysis, however, are not conclusive about the progression of PD in early to advanced patients. No major biomarker of ROIs functional connectivity, nor any progressive dissociation were observed between Prodromal and PD cohorts.

4.3.2 Brain Network Topology Analysis

Figure 8 shows the distributions of seven global network features extracted from resting-state functional connectivity across tree graphs generators: Absolute Thresholding (top row), Fixed Density (middle row), and Orthogonal Minimum Spanning Tree (OMST; bottom row). Each boxplot compares the feature values across the Control, Prodromal, and Parkinson groups using the Shen 268 parcellation.

Analysing Figure 8, it is possible to identify a consistent decrease trend in local efficiency that follows the progression of PD across all three networks generators. This finding is supported by other studies that found similar trend in local efficiency in PD patients (GöTTLICH *et al.*,

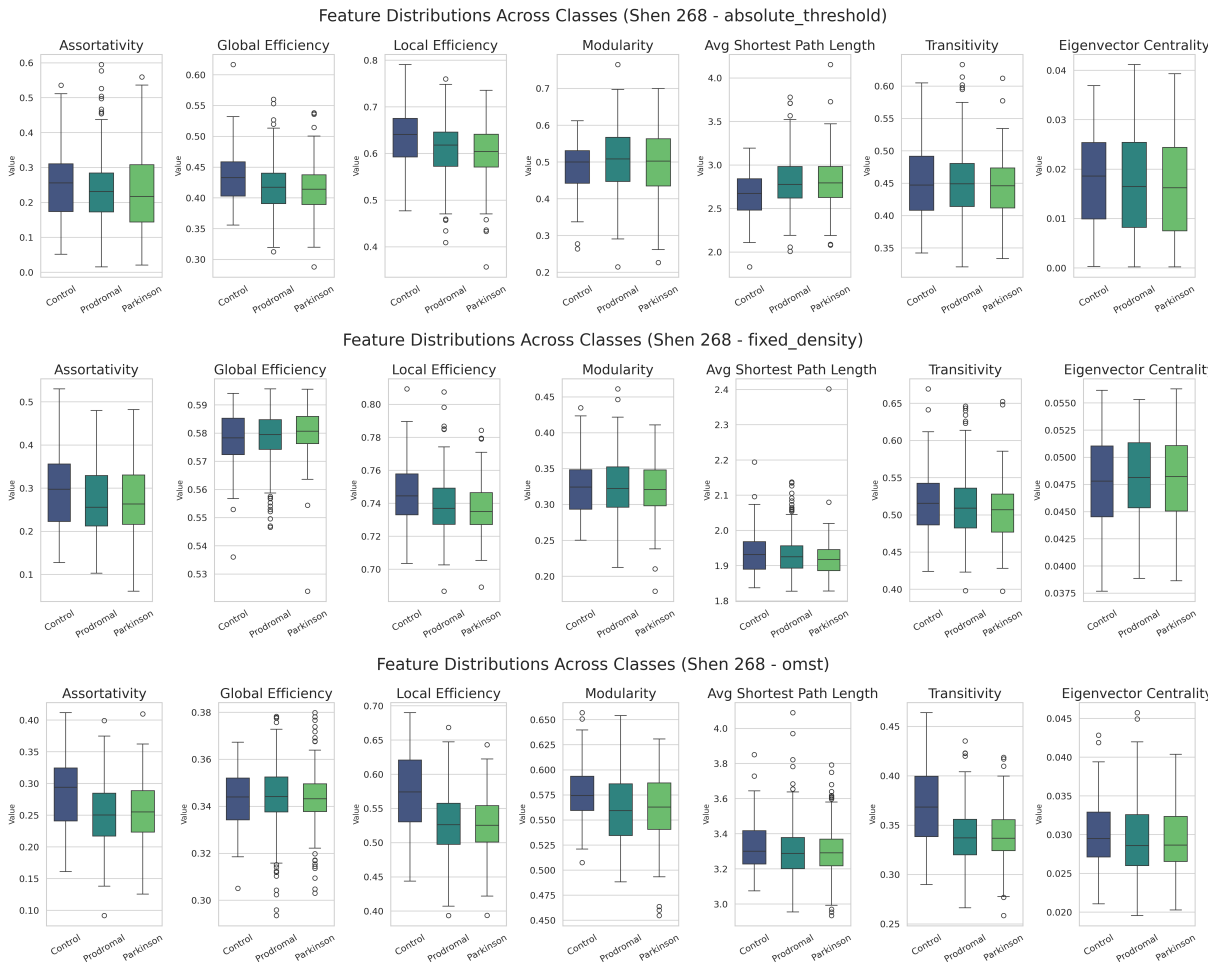

Figure 8 – Distributions of complex network features

2013) (SKIDMORE *et al.*, 2011). This decrease in the local efficiency indicates an impaired global integration of neural information in PD patients. Other measures, however, do not show a conclusive trend.

Our results suggest that graph metrics might be used to identify and track neurodegenerative diseases, however more studies will be needed to evaluate utility of this network topological type of analysis.

4.3.3 Most Discriminative Connections

To uncover the most diagnostically informative functional connections in distinguishing Parkinson's disease from control conditions, we analyzed global SHAP (SHapley Additive exPlanations) values derived from a Random Forest classifier trained on the full set of pairwise resting-state connectivity features extracted from the Shen 268-node atlas. Each feature represented the functional connectivity between a specific pair of regions of interest (ROIs), and SHAP values were used to quantify the individual contribution of each feature to the model's decision-making process across all classes.

A network-level summary was generated by grouping SHAP values according to the resting-state network assignments of the ROIs. The aggregated mean absolute SHAP values between network pairs are visualized in Figure 9. This symmetric heatmap highlights both within- and between-network contributions to the classification model. The most influential interactions were primarily intra-network within the Subcortical-Cerebellum, as well as inter-network connections involving Subcortical-Cerebellum and Motor systems.

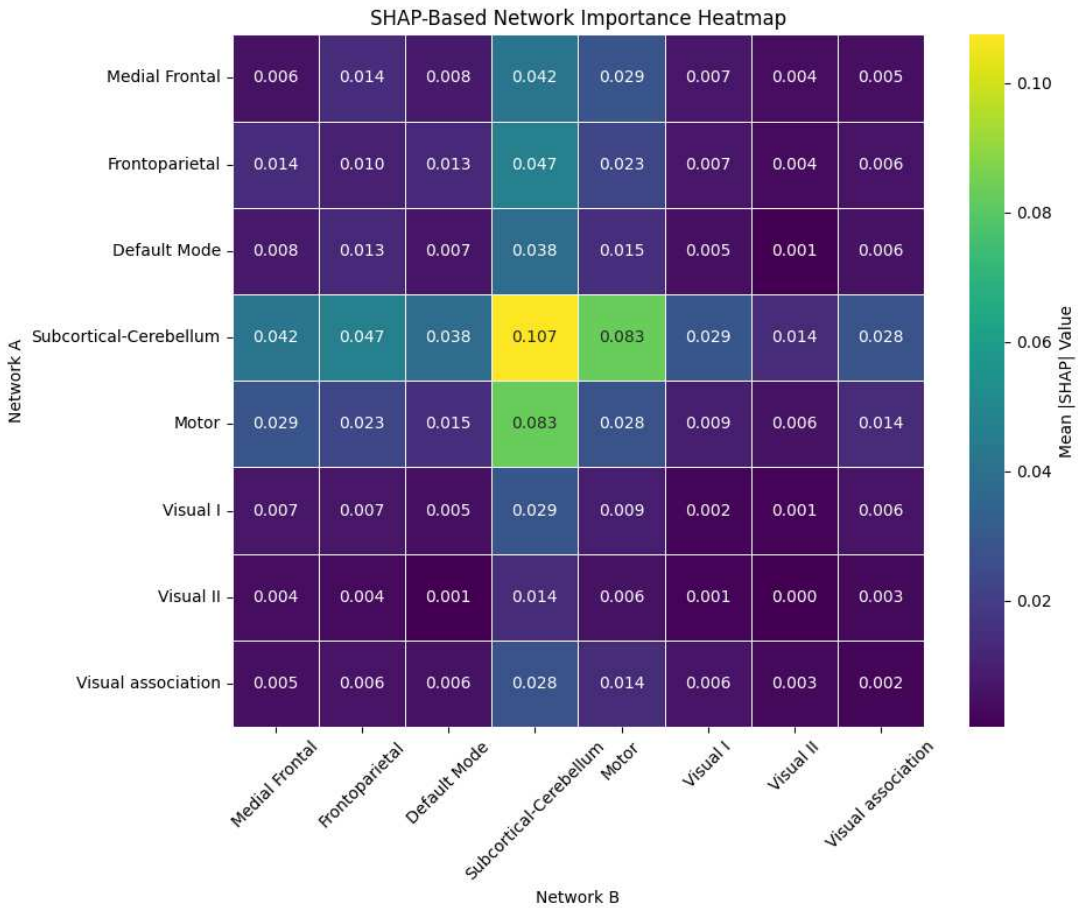

Figure 9 – Heatmap of aggregated mean absolute SHAP values between network pairs

Figure 9 reveals a similar pattern to the findings of subsection 4.3.1. This indicates that the machine learning model is taking into consideration connectivity alterations between specific brain networks that are also evident in the group-level statistical differences. Most importantly, it shows that the model in not learning from noise. In particular, the model attributes higher discriminative importance to connections involving the Subcortical-Cerebellum and Motor networks, that were consistently highlighted as altered in the connectivity difference maps. This convergence strengthens the interpretability of the classifier by linking its predictive behaviour to known neurobiological disruptions in Parkinson's Disease as mentioned in Section 4.3.1.

Chapter 5

CONCLUSION

This research was motivated by the urgent need for quantitative, reproducible, and interpretable methods for diagnosing Parkinson's Disease (PD). Current clinical diagnoses are often subjective, achieving around 80% accuracy in later stages but as low as 26% in early, untreated cases. This diagnostic gap highlights a critical need for objective biomarkers. This project aimed to develop an automatic diagnostic methodology by integrating machine learning and resting-state fMRI connectomics, with a specific goal of not only classifying patients but also providing interpretable biological insights.

Our results demonstrated a clear distinction in the diagnostic power of different feature-engineering approaches. A modest classification performance was achieved using abstracted, graph-theoretic network measures (best test F1-macro score of 0.478). In contrast, a superior approach utilizing the high-dimensional, ROI-to-ROI functional connectivity matrix as a direct feature set achieved a test F1-macro score of 0.665 ± 0.064. This 39% relative improvement, obtained using a Logistic Regression classifier on the functionally-derived Shen 268 atlas, supports the hypothesis that raw connectivity patterns hold the most critical biomarkers.

Beyond classification, this work identified interpretable neurobiological markers. Both statistical group analysis and machine learning interpretability converged on the same key finding: PD pathophysiology is overwhelmingly associated with connectivity disruptions centered on the Subcortical-Cerebellar (SC) network. We identified a widespread pattern of decreased connectivity, both within the SC network (involving the basal ganglia, thalamus, and cerebellum) and between the SC network and other major cortical networks. This finding provides quantitative support for the characterization of PD as a "disconnection syndrome". A consistent decrease in local efficiency was also observed in line with disease progression, suggesting impaired global information integration.

A key contribution of this thesis was ensuring this classifier was not a "black box." SHapley Additive exPlanation (SHAP) analysis confirmed that the model's most influential features were these same biologically-relevant connections within the SC network and between the SC and Motor networks. This convergence demonstrates that the model learned a meaningful neurofunctional signature of the disease, rather than spurious noise, which is essential for building clinical trust and moving toward explainable decision-support.

Despite these successes, we must acknowledge several limitations. While the best model's

performance is promising, a 0.665 F1-macro score is not yet sufficient for deployment as a standalone clinical tool. More importantly, while our model could effectively distinguish patients (Prodromal and PD) from Healthy Controls, we did not identify clear biomarkers or achieve strong classification performance in differentiating the stages of the disease (i.e., Prodromal vs. established PD). This remains a significant challenge, as the neurodegenerative changes between these two stages are likely more subtle.

Future work should focus on overcoming this challenge. This could involve exploring methods such as deep learning or testing the effect of different preprocessing pipelines. Furthermore, integrating fMRI data with other modalities, such as clinical data and genetic markers could provide a more powerful predictive model. Finally, the promising Shen 268 atlas + Logistic Regression model should be validated on larger and independent datasets to confirm its generalizability.

In conclusion, this research successfully developed a quantitative and interpretable framework for PD diagnosis, confirmed the superiority of raw functional connectivity data, and provided robust biological insights into the central role of subcortical-cerebellar disconnection in Parkinson's disease.

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